INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **BCA** Semester: **Fourth**

Stream: **BCA**

Paper Code: **BCACC403**

Paper Name: **Operating Systems**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1.	What is multiprogramming?	**2**
2.	What is a process?	**2**
3.	Write one necessary condition to occur a deadlock.	**2**
4.	What is init process?	**2**
5.	What does PCB contain?	**2**
6.	What is main demerit of is LRU page replacement algorithm?	**2**
7.	When does starvation occur?	**2**
8.	What is a semaphore?	**2**
9.	How free-space is managed using bit vector implementation?	**2**
10.	What does sleep () do?	**2**
11.	What is the difference between Internal and External fragmentation?	**2**
12.	What is a safe state?	**2**
13.	What is mutual exclusion?	**2**

14. Mention the objectives of File Management System. 2

15. What is the difference between seek and latency time? 2

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Explain four necessary conditions for deadlock to occur 5

17. Present Banker's Algorithm and explain. 5

18. Describe the differences among short-term, medium-term, and long-term scheduling. 5

19. Consider the following table and evaluate the average turnaround time and average waiting time for SJF scheduling algorithm. 5

Process Name	Arrival Time	Processing Time	Priority	Time Quantum
A	0	3	2	2
B	2	6	1	2
C	4	4	3	2
D	6	5	4	2
E	8	2	5	2

20. Design a Resource Allocation Graph which has the presence of Deadlock. 5

21. A system uses 3 page frames for storing process pages in main memory. Assume that all the page frames are initially empty. Calculate the total number of page faults for LRU while processing the page reference string given below 5
4 , 7, 6, 1, 7, 6, 1, 2, 7, 2

22. Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending requests, in FIFO order is: 5
86,1470,913,1774,948,1509,1022,1750,130
Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms?
    a. FCFS
    b. SSTF

23. Consider a logical address space of 32 pages with 1,024 words per page, mapped onto a physical memory of 16 frames. 5

a. How many bits are required in the logical address?
b. How many bits are required in the physical address?

24. What are the main advantages and disadvantages of using a spinlock for synchronization? **5**

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Consider the following table and evaluate the average turnaround time and average waiting time for FCFS and SJF **10**

Process Name	Arrival Time	Processing Time	Priority	Time Quantum
A	0	3	2	3
B	2	6	1	3
C	4	4	3	3
D	6	5	4	3
E	8	2	5	3

26. Assume arrival order is: P1, P2, P3, P4, P5 at time 0, 1, 2, 3, 4 respectively and a smaller priority number implies a higher priority. Draw the Gantt charts for preemptive and non-preemptive priority scheduling. Calculate Average Turnaround Time and Average Waiting Time. **10**

27. Consider the following snapshot of a system: **10**

	Allocation	Max	Available
	A B C D	A B C D	A B C D
P_0	0 0 1 2	0 0 1 2	1 5 2 0
P_1	1 0 0 0	1 7 5 0	
P_2	1 3 5 4	2 3 5 6	
P_3	0 6 3 2	0 6 5 2	
P_4	0 0 1 4	0 6 5 6	

28. Consider the Pages referenced by the CPU in the order are 6, 7, 8, 9, 6, 7, 1, 6, 7, 8, 9, 1, 7, 9, 6 with 3 page frames. Apply the LRU and Optimal Algorithm to find page fault. **10**

29. A real-time embedded system uses fixed-partition memory allocation but suffers from external fragmentation. Propose and analyse the impact of paging and segmentation on memory utilization. Compare and contrast their effectiveness in handling fragmentation in such real-time systems. **10**

30. Given memory partition of 100K, 500K, 200K, 300K, and 600K in order, how would each of the First-fit, Best-fit and Worst-fit algorithms place the processes of 212K, 417K, 112K and 426K in order? Which algorithm makes the most efficient use of memory? Show the diagram of memory status in each case. **10**

31. What are points to be consider in file system design? Explain linked list allocation & index allocation in detail. **10**

32. A disk scheduling system receives the following request queue (in cylinder numbers): 98, 183, 37, 122, 14, 124, 65, 67
If the initial head position is 53, calculate the total head movement using FCFS, SSTF, and SCAN algorithms. **10**

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